

1.1

dokaŹte, v komplexnĚm V s $\langle, \rangle$
 platĚ $\forall u, v \in V$. MĚjme

$$\text{vĚme } \overline{\langle u, v \rangle} = \langle v, u \rangle$$

$$\text{a } \|x\|^2 = \langle x, x \rangle$$

$$\Rightarrow \langle u+v, u+v \rangle = (\|u+v\|^2) =$$

$$= \langle u, u \rangle + \langle u, v \rangle + \langle v, u \rangle + \langle v, v \rangle =$$

$$= \|u\|^2 + \langle u, v \rangle + \overline{\langle u, v \rangle} + \|v\|^2 = \|u\|^2 + \|v\|^2 + 2\operatorname{Re}(\langle u, v \rangle)$$

$$\Rightarrow \langle u-v, u-v \rangle =$$

$$= (\|u-v\|^2) = \langle u, u \rangle$$

$$- \langle u, v \rangle - \langle v, u \rangle$$

$$+ \langle v, v \rangle = \|u\|^2 - \langle u, v \rangle -$$

$$- \overline{\langle u, v \rangle} + \|v\|^2 =$$

$$\|u\|^2 + \|v\|^2 - 2\operatorname{Re}(\langle u, v \rangle)$$

$\implies$

$$\|u+v\|^2 - \|u-v\|^2 =$$

$$= 4\operatorname{Re}(\langle u, v \rangle)$$

$\implies$

$$\operatorname{Re}(\langle u, v \rangle) = \frac{\|u+v\|^2 - \|u-v\|^2}{4}$$

$\operatorname{Im}(\langle u, v \rangle):$

$$\langle u-iv, u-iv \rangle = \|u-iv\|^2 =$$

$$= \dots = \|u\|^2 + \|v\|^2$$

$$- i\langle u, v \rangle$$

$$+ i\overline{\langle u, v \rangle}$$

$$\left(\text{dla twierzenia: } \overline{\langle u, v \rangle} = \langle u, v \rangle - 2i \operatorname{Im}(\langle u, v \rangle) \right)$$

$$\dots = \|u\|^2 + \|v\|^2 + 2 \operatorname{Im}(\langle u, v \rangle)$$

$$\|u + iv\|^2 =$$

$$= \|u\|^2 + \|v\|^2 +$$

$$-i \overline{\langle u, v \rangle} + i \langle u, v \rangle =$$

$$= \|u\|^2 + \|v\|^2 - 2 \operatorname{Im}(\langle u, v \rangle)$$

$$\Rightarrow \operatorname{Im}(\langle u, v \rangle) = \frac{\|u - iv\|^2 - \|u + iv\|^2}{4}$$